

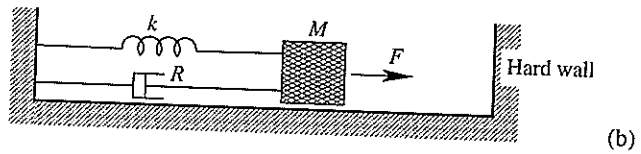
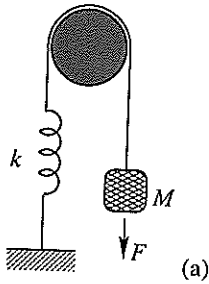
IISER-Kolkata
Classical Mechanics
Mid-semester supplementary examination

Ray

25/09/2024, 3 PM to 4.30 PM

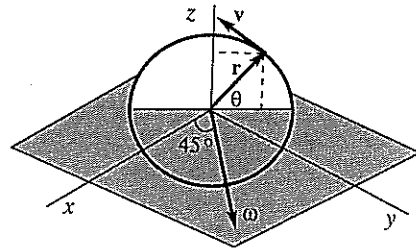
Answer question 1 and any two from the rest.

1. (a) Why does the moment of inertia of a rigid body have to be represented by a tensor? 2
- (b) What is a virtual displacement? 2
- (c) What is the 'Least Action Principle'? 2
- (d) Show that for a Hamiltonian system,
 $\dot{q}_j = \{q_j, H\}$ and $\dot{p}_j = \{p_j, H\}$ 4
2. Define 'holonomic constraint'. Are the following systems holonomic or non-holonomic?
 Write down the constraint equations (or inequations) for each system. (7+8)



3. A particle is rotating in a vertical plane. Its angular velocity vector lies in the xy plane and makes an angle of 45° with the x and y axes.

- (a) Find the vector $\vec{r}$ (3)
- (b) Find the vector $\dot{\vec{r}}$ by differentiating $\vec{r}$ (3)
- (c) Find the cross product $\vec{\omega} \times \vec{r}$ (3)
- (d) Find the angular momentum of the particle. (6)



4. Obtain the Lagrangian equations of motion for the system

(15)

